Perception Without Optics: AI Vision as a Post-Sensory Epistemology

Introduction: The Post-Sensory Condition

The contemporary technological landscape is haunted by a profound philosophical paradox: the emergence of systems that "see" without sensory organs. This phenomenon, which can be termed "perception without optics," marks a fundamental epistemological rupture. For centuries, Western thought has tethered knowledge of the world—epistemology itself—to the sensory apparatus of an embodied observer. From empiricism's reliance on sense-data to phenomenology's grounding in the lived body, the act of seeing has been predicated on the physical interaction of light, lens, and retina. Today, generative artificial intelligence, particularly diffusion models, challenges this foundational assumption. These systems can produce images of staggering photorealistic quality, simulating the intricate physics of light, the nuances of depth, and the specific artifacts of lens behavior, all without ever interacting with a single photon or occupying a point in physical space.[1]

This disembodied vision, a form of camera awareness detached from any camera, forces a critical re-evaluation of our most basic concepts. When a disembodied algorithm can generate a "photograph" that is visually indistinguishable from one produced by a physical camera, what becomes of the photograph's traditional claim to indexical truth—its status as a physical trace of reality?.[3] When a system can simulate depth perception by generating depth maps and parallax effects from statistical patterns, how does this reconfigure our understanding of experience, which has been so intimately linked to embodiment and action?.[5] And when a model's "understanding" of the world is not derived from sensory feedback but from the statistical regularities of a billion-image dataset, what does this imply about the nature of cognition itself?.[7]

This report conducts a layered investigation into the ontological status and epistemological implications of this new post-sensory condition. It argues that AI-generated imagery is not merely a new form of digital manipulation but represents a distinct ontological category: the post-sensory image. To analyze this phenomenon, the report moves through six interconnected inquiries. Part I traces a genealogy of realism, charting the historical shift from the photograph's indexical truth-claim to the AI's algorithmic plausibility, drawing on the media theory of Vilém Flusser and Jonathan Crary. Part II offers a technical and conceptual analysis of diffusion models, demystifying their mechanics to posit the "latent space" as an abstract perceptual field governed by statistical priors. Part III stages a phenomenological confrontation, contrasting the AI's disembodied coherence with the embodied perception described by Maurice Merleau-Ponty and the predictive processing framework of Andy Clark, thereby locating the precise nature of the philosophical rupture. Part IV examines the role of cultural conditioning and viewer complicity in granting these images epistemic legibility, using edge-case prompts to probe the limits of their simulated realism. Part V reframes AI cognition as a form of "alien intelligibility," using non-human animal vision as an analogical tool to de-center anthropocentric evaluations. Finally, Part VI proposes a path toward a more rigorous future epistemology for AI vision, formulating testable hypotheses and a speculative metric for "epistemic fidelity"

that can audit the physical plausibility of these post-sensory images. Through this comprehensive analysis, this report aims to construct a robust philosophical framework for understanding a world increasingly populated by images that see without eyes and represent without referents.

Part I: From Index to Algorithm: A Genealogy of Realism

The emergence of AI-generated imagery, with its capacity to synthesize photorealistic scenes from textual prompts, appears to represent a radical break from the history of visual representation. However, a genealogical analysis reveals that this development is not a complete departure but rather the logical, if extreme, culmination of a process of abstraction inherent to photography itself. The transition from the camera's mechanical eye to the algorithm's statistical gaze marks a definitive shift from an epistemology of **indexical realism** to one of **algorithmic realism**. In the former, an image's truth-claim was tethered, however tenuously, to a physical trace of light; in the latter, truth is supplanted by statistical plausibility, and evidence is replaced by procedural proof-of-work.[9] Understanding this transition requires excavating the philosophical foundations of the photographic apparatus and the historical construction of the observer.

The Photographic Apparatus and the Constructed Observer

The philosopher Vilém Flusser, in his seminal work *Towards a Philosophy of Photography*, provides a crucial framework for deconstructing the apparent transparency of the photographic image. Flusser argues that the camera is not a neutral tool for capturing reality but an "apparatus" that imposes its own "program" on the world.[11] The photographer, far from being a sovereign creator, is a "functionary" of this apparatus, a player in a game whose rules are predetermined by the camera's design.[11] The act of taking a photograph is therefore not a passive recording but a conceptual act of encoding the world into a "technical image"—an image that is fundamentally an abstraction derived from scientific texts (theories of optics, chemistry, etc.).[12] Flusser posits that this shift from traditional images (like paintings) to technical images marks a profound cultural turning point, moving consciousness from a linear, historical mode (associated with text) to a magical, two-dimensional one (associated with the surface of the image).[14] This perspective dismantles the naive belief in photographic objectivity, revealing the camera as a "black box" that structures reality according to its internal logic long before the advent of AI.

Complementing Flusser's analysis of the apparatus, media theorist Jonathan Crary's work in *Techniques of the Observer* deconstructs the myth of a stable, ahistorical observer. Crary argues that the most significant transformation of vision in modernity occurred *before* the invention of photography, during the early 19th century.[16] This period witnessed the collapse of the classical model of vision, symbolized by the camera obscura, which posited a clear separation between the observing subject and the external world.[17] It was replaced by models of "subjective vision," where vision was relocated within the body as a physiological event, subject to scientific study and new forms of discipline and standardization.[17] For Crary, optical devices like the stereoscope, which predated photography, were "sites of knowledge and power that operate directly on the body of the

individual," producing a new kind of observer whose perception was already mediated and constructed.[17]

From this perspective, AI image generation represents the final and most decisive "uprooting of vision from the stable and fixed relations incarnated in the camera obscura".[17] It creates a visuality that is entirely "severed from a human observer," relocating perception to a "cybernetic and electromagnetic terrain" where abstract visual and linguistic elements are processed and exchanged globally.[19] The historical process described by Flusser and Crary demonstrates that the observer and the image were always constructs of technology and power. The belief in a pure, unmediated visual truth was a cultural convention, not a physical fact. AI does not betray this history; it fulfills it by severing the final, tenuous link to a physical referent, making the constructed nature of the image explicit.

The Split: From Indexical to Algorithmic Realism

The authority of the photograph throughout the 19th and 20th centuries was founded on the principle of **indexical realism**. As defined in semiotics by Charles Sanders Peirce and applied to photography by theorists like Roland Barthes, an indexical sign bears a direct, physical, causal relationship to its object.[4] A photograph was considered an index because it was formed by the physical imprint of light rays reflecting off a real-world object and striking a photosensitive surface.[4] This "direct trace" of the world gave photography its immense power as a medium of evidence, memory, and truth.[3] As Susan Sontag noted, the photograph's authority stemmed from this physical touch, which established its relationship to the past and to fact.[21]

However, this indexical authority was never absolute. As Eyal Weizman's analysis of low-resolution satellite imagery shows, a single pixel can be profoundly ambivalent, serving as "evidence of a drone attack or proof that it never happened".[21] The index could obscure as much as it revealed, hiding ambiguity within its claim to facticity. The history of photography is also rife with manipulation, from staging scenes to combining negatives, demonstrating that the "truth" of the photograph has always been a negotiated, rather than given, quality.[22]

The advent of AI generative models precipitates a definitive break from this paradigm, instituting what can be termed **algorithmic realism**. This concept, articulated by thinkers like Phil Chang and Ben Green, describes a mode of representation where realism is no longer tied to depiction or indexicality but to function, network logic, and statistical plausibility.[9] In the world of AI-generated images, realism is decoupled from its traditional foundations and becomes a "mere stylistic effect".[3] These images are not representations of reality but "statistical reconstructions" of patterns found in vast datasets.[3] They are, as Krešimir Purgar suggests, a form of "proxy-real"—substitutes for reality rather than representations of it.[3]

The "truth" of an AI image, therefore, is not located in its correspondence to a physical event but in the integrity of the computational process that created it. As one analysis of the first algorithmic image of a black hole argues, we have already transitioned to a standard of representation based on "algorithmic proof-of-work"—that is, on *how* an image was created, rather than what it looks like or what it points to.[10] An AI image functions not indexically but "relationally, as coordinates in a vast conceptual space".[3] This new realism is not more truthful than photography; it is simply more dominant, structuring our understanding of the world with greater immediacy and efficiency.[9] The "crisis of realism" prompted by AI is not, therefore, the introduction of falsehood into a world of visual truth, but the

final, explicit acknowledgment of a process of abstraction that was present from photography's very inception.

Part II: The Latent Space as an Abstract Perceptual Field

To comprehend the epistemological rupture presented by AI vision, one must move beyond metaphor and engage with the technical architecture that underpins it. Unlike a camera, which captures photons, a diffusion model generates images through a purely mathematical process of statistical inference. Its "vision" is not optical but probabilistic, and its perceptual field is not physical space but a high-dimensional, abstract manifold known as the **latent space**. Within this space, the model's "knowledge" of the world exists as a set of learned statistical regularities, or **image priors**. The act of "seeing," for a diffusion model, is the process of navigating this abstract space to reconstruct a plausible image from pure noise, guided by the powerful constraints of these priors.

The Mechanics of Diffusion: From Data to Noise and Back

The operational logic of a diffusion model is inspired by non-equilibrium thermodynamics and can be understood as a two-part process.[24]

First, the **forward diffusion process** is a fixed, non-learned procedure that systematically degrades an image from the training data. This is modeled as a Markov chain, where at each timestep t in a sequence of T steps, a small amount of Gaussian noise is added to the image from the previous step, x_{t-1}.[26] The process is defined by a variance schedule,

β_t, which controls the amount of noise added at each step. The state of the image at time t can be expressed as:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t} x_{t-1}, \beta_t I)$$

where N is a Normal (Gaussian) distribution. After T steps (often 1000 or more), the original image x_0 is transformed into an almost pure isotropic Gaussian noise distribution, x_T.[26] This process effectively converts a complex, unknown data distribution (the distribution of all possible images) into a simple, tractable one (a standard normal distribution).[26]

Second, the **reverse diffusion process** is where the learning occurs. The model, typically a neural network, is trained to reverse the forward process step-by-step.[24] Its goal is to learn the reverse transition probability

$p_\theta(x_{t-1} | x_t)$, where θ represents the learnable parameters of the network. In practice, this is achieved by training the network to predict the noise ϵ that was added to the image at each timestep t.[30] The model takes the noisy image

x_t and the timestep t as input and outputs a prediction of the noise, $\epsilon_\theta(x_t, t)$. The loss function is typically a simplified objective that minimizes the mean squared error between the actual noise ϵ and the predicted noise ϵ_θ:

$$L_{simple}(\theta) = \mathbb{E}\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right|^2

\right]$$
where αt=1−βt and α¯t = \prod{s=1}^{t} \alpha_s. By learning to predict and subtract this noise iteratively, the model can start with a random sample of pure noise xT and gradually denoise it back into a coherent image that resembles the training data.[30]

The U-Net as the Engine of Reconstruction

The neural network architecture most commonly used for this denoising task is the **U-Net**.[32] Originally developed for biomedical image segmentation, the U-Net's design is exceptionally well-suited for diffusion models.[34] Its architecture is symmetrical, consisting of a

contracting path (encoder) and an **expansive path (decoder)**, giving it a characteristic "U" shape.[34]

The contracting path progressively downsamples the input image through a series of convolutional and pooling layers, capturing high-level, contextual features while reducing spatial resolution.[34] The expansive path then upsamples these feature maps, systematically reconstructing the image's spatial details. The crucial innovation of the U-Net is the use of

skip connections, which concatenate feature maps from the contracting path directly with the corresponding layers in the expansive path.[34] These connections allow the decoder to leverage both high-level semantic context from the deeper layers and fine-grained, high-resolution feature information from the earlier layers. This multi-resolution analysis makes the U-Net highly effective at predicting the noise present in an image at various scales, from broad patterns to minute textures.[36]

Latent Space and Image Priors: The Locus of "Knowledge"

Operating directly on high-resolution images is computationally expensive. To address this, state-of-the-art models like Stable Diffusion are **Latent Diffusion Models (LDMs)**.[38] In an LDM, the diffusion process does not occur in the high-dimensional pixel space. Instead, a pre-trained

Variational Autoencoder (VAE) is used to first compress the image into a much smaller, lower-dimensional **latent space**.[32] The VAE consists of an encoder that maps the image to a latent vector and a decoder that reconstructs the image from that vector.[41] This latent representation is designed to capture the essential semantic content of the image while discarding redundant pixel-level information.[38] The entire forward and reverse diffusion process then takes place within this computationally efficient latent space.[39]

This latent space is the AI's true perceptual field. It is not a geometric representation of the physical world but a high-dimensional abstract manifold where concepts, objects, and styles are organized based on their statistical co-occurrence in the training data.[32] Proximity in this space corresponds to semantic similarity. The act of "seeing"

or generating an image is a guided traversal of this space, a path from a random point (noise) to a specific region corresponding to the desired concept (e.g., "a cat sitting on a mat").

The model's ability to perform this traversal and generate coherent, realistic images is entirely dependent on the powerful **image priors** it learns during training.[43] An image prior is the learned, implicit knowledge of the underlying statistical distribution of the visual world as represented in the training data.[43] The model does not possess an explicit, symbolic understanding of what a face is. Instead, through exposure to millions of examples, it develops an incredibly strong prior for the statistical regularities that define a face: the typical arrangement of eyes, nose, and mouth; the texture of skin; the way light reflects on its surface.[45] The generative process is a form of Bayesian inference where this powerful prior is used to impose structure on a random input.

This reveals a fundamental distinction: human vision begins with an external stimulus—light from the world impacting the retina. AI vision begins with an internal state—a random vector of noise in a latent space.[41] The process is not one of

capturing or *registering* an external reality, but of *imposing* a learned, internal structure onto randomness. The U-Net does not "see" a noisy image and recognize objects within it; it performs a mathematical operation, predicting a noise map based on the current latent state and a given timestep, guided by the immense statistical knowledge encoded in its parameters.[30] The resulting image's fidelity is not a measure of its correspondence to a unique physical referent, but a measure of its conformity to the learned probability distribution of the training data. It is a perfect instantiation of Flusser's "technical image," which abstracts not from the world itself, but from the texts, data, and algorithms that model it.[12]

Part III: The Rupture of Embodiment: Phenomenological Inquiries

The technical analysis of diffusion models reveals a system that generates visual plausibility through statistical inference, entirely detached from sensory input. This disembodied "vision" represents a profound philosophical rupture, one that can be precisely located by contrasting it with phenomenological accounts of human perception. By placing the AI's latent coherence in dialogue with Maurice Merleau-Ponty's concept of the **lived body** and Andy Clark's theories of **predictive processing** and the **extended mind**, we can move beyond a technical description to an ontological one. This contrast reveals that AI "perception" is not a deficient or nascent form of human perception but a different category of being altogether, one defined by the absence of the embodied, action-oriented engagement that constitutes our being-in-the-world.

Merleau-Ponty: The Primacy of the Lived Body (Le Corps Propre)

For the French phenomenologist Maurice Merleau-Ponty, perception is the primordial mode of our access to the world, and it is fundamentally an embodied activity.[48] He vehemently rejects the Cartesian dualism of mind and body, arguing that we are not disembodied minds piloting a mechanical body; rather, our consciousness is

inextricably interwoven with our physical form.[6] We are not "thinking beings that happen to have bodies," but "embodied beings".[51] The

lived body (*le corps propre*) is not an object among other objects but our very means of "having a world," the active mediator that generates meaning through its sensory and motor interactions.[49]

This embodied perception is structured by two key concepts. The first is the **body schema** (*schéma corporel*), a pre-reflective, dynamic, and practical sense of the body's posture, capabilities, and orientation in space.[53] It is not a conscious representation but an operational "I can" that allows us to navigate our environment seamlessly, such as moving through a doorway without consciously comparing our body's width to the opening.[53] The second is the

intentional arc, which "projects round about us our past, our future, our human setting, our physical, ideological and moral situation".[55] This arc, subtended by the body schema and our acquired habits, is what allows the world to show up for us as a field of meaningful possibilities, or "affordances." A hammer affords hammering, and a chair affords sitting, not because of their objective properties alone, but because our bodies have the capacity and learned skill to interact with them in these ways.[55] For Merleau-Ponty, perception and action exist in a state of "circular causality"; we perceive the world in terms of what we can do in it, and our actions, in turn, shape our perceptions.[6]

An AI, by its very nature, suffers from a profound **embodiment deficit**.[56] Lacking a lived body, it has no body schema and thus no genuine intentional arc.[52] Its "understanding" of a chair is not an embodied possibility for sitting but a statistical correlation between the token "chair" and a cluster of pixels in its high-dimensional latent space.[58] It has no pre-reflective "grip" on the world, no history of motor interaction that gives objects their meaning. This absence is not a mere technical limitation to be overcome but a fundamental ontological distinction.

Case Study: The Simulation of Depth Perception

The AI's simulation of depth perception provides a powerful case study for this phenomenological rupture. Diffusion models can generate images with a convincing sense of three-dimensionality.[60] They achieve this not by perceiving space but by learning to simulate the visual cues associated with depth. For instance, models like Marigold can be fine-tuned to produce highly accurate

depth maps from a single 2D image.[5] This depth map—a grayscale image where pixel intensity corresponds to distance—is then used to algorithmically generate optical effects. The

parallax effect, which creates the illusion of stereoscopic vision, is simulated by calculating a pixel shift for left and right eye views based on the depth value and a predefined interpupillary distance.[5] Similarly, the

bokeh effect (shallow depth of field) is created by applying a Gaussian blur whose intensity is directly proportional to a pixel's value in the depth map, mimicking the way a camera lens blurs the background and foreground.[5]

From a phenomenological perspective, this process is a simulation of visual artifacts, not an experience of depth. For a human, depth is not a calculated property derived from a 2D map. It is a lived, multi-sensory experience of

spatiality, of being-in-the-world, structured by our body's potential for movement.[6] We perceive the "unseen side" of an object not through intellectual synthesis but through a "practical synthesis"—the knowledge that we

can move around it and touch it.[59] The AI masters the statistical cues of depth as they appear in two-dimensional photographs, but it lacks the embodied, action-oriented foundation that gives depth its meaning for us.

Andy Clark: Predictive Processing and the Extended Mind

The work of philosopher and cognitive scientist Andy Clark offers another lens through which to analyze this disembodiment. Clark's theory of **predictive processing** posits the brain as a "prediction machine".[63] In this model, the brain does not passively receive sensory input. Instead, it actively generates top-down predictions about what it expects to sense, based on its internal generative model of the world. These predictions are then compared against the actual bottom-up sensory information. Any mismatch results in a "prediction error," which is used to update the internal model.[63] Perception, in this view, is a form of "controlled hallucination," where our expectations constantly shape our experience.[67]

A diffusion model can be seen as a powerful prediction engine; its core function is to predict the noise that needs to be removed from a noisy input.[30] However, it is a predictor disconnected from a world. It lacks the crucial feedback loop of bottom-up sensory input and prediction error correction that grounds the human brain's generative model in reality.[65] Its "controlled hallucination" is unconstrained by sensory data, governed solely by its internal statistical priors learned from a static dataset. It generates predictions without ever having its predictions challenged by an external world.

Furthermore, Clark's **extended mind thesis** proposes that cognitive processes can literally extend beyond the brain and body to include external tools.[69] A notebook used to store information becomes a part of one's memory system, functioning as an external cognitive resource.[69] While an AI can certainly function as such a tool, its disembodiment prevents it from becoming part of a truly "coupled system" in the way a blind person's cane becomes a seamless extension of their sensory body, mediating a direct, tactile relationship with the world.[6] The cane is integrated into the user's body schema; the AI remains an external oracle.

The convergence of these phenomenological and cognitive science perspectives leads to a clear conclusion. The AI's ability to simulate perception is not a nascent form of being but an ontologically distinct process. Merleau-Ponty's framework shows that human perception is defined by the body's active, skillful engagement with a meaningful world. Clark's framework highlights that even if we model cognition as prediction, this process is meaningless without the corrective feedback loop of sensory experience. The AI's "perception" is a system of pure simulation, a coherent hallucination generated from statistical priors, entirely detached from the existential, embodied condition that defines life and consciousness.

Part IV: The Eye of the Beholder: Epistemic Legibility and Viewer Complicity

The remarkable success of AI-generated images in achieving photorealism hinges not only on technical sophistication but also on a crucial, often unexamined, factor: viewer complicity. Our ability to "read" these synthetic images as photographically true is a product of deep-seated cultural conditioning. We grant them epistemic authority because they have mastered the aesthetic conventions that we have learned to associate with photographic truth. This section argues that we are active participants in the illusion of algorithmic realism. By probing the model's limits with edge-case prompts that violate these conventions, we can expose its underlying logic as statistical rather than physical. This process reveals the brittleness of its realism and highlights the ethical stakes of its deployment, particularly the potential for what can be termed **generative algorithmic epistemic injustice**.

The "Photographic Look" and Cultural Conditioning

AI image generators are adept at producing what can be called the "photographic look".[2] This aesthetic is not a neutral reflection of reality but a collection of stylistic conventions that have become synonymous with professional photography. These include a shallow depth of field, often referred to as "bokeh," which isolates a sharp subject against a pleasingly blurred background; specific color grading palettes, such as a "bronzy, very warm look"; and dramatic, often cinematic, lighting.[2] These are the visual signifiers of quality and authenticity that saturate our visual culture, from advertising to social media.[73]

Our acceptance of these signifiers as markers of truth is a result of cultural conditioning.[75] For over a century, photography's authority was grounded in its indexical promise—the belief that it was a direct, physical trace of reality.[4] Despite a long and well-documented history of photographic manipulation, from the earliest composite landscapes to modern digital retouching, the mantra that "the camera never lies" persists in the popular imagination.[22] AI image generators exploit this ingrained trust. They do not produce an indexical trace, but they expertly simulate the

style of indexicality.[3] Realism, in this context, is transformed from an epistemic standard into a selectable aesthetic option.[3]

Studies on viewer perception reveal our complicity in this process. When viewers are unaware of an image's origin, they often show no aversion to AI-generated content and may even prefer its polished, hyper-realistic aesthetic.[78] However, the mere knowledge that an artwork was created by AI can induce a negative bias, causing viewers to look at it for shorter durations, suggesting an implicit prejudice against machine creativity even when the visual output is aesthetically pleasing.[78] This demonstrates that our judgment of an image's "truth" or "authenticity" is heavily influenced by contextual framing, not just its visual content.

Epistemic Injustice and Representational Harm

The convincing plausibility of AI images comes with significant ethical costs. Because diffusion models are trained on vast, uncurated datasets scraped from the internet, they inevitably inherit and amplify the biases present in that data.[73] This leads to what can be defined as

generative algorithmic epistemic injustice: the production of content that systematically misrepresents or erases the experiences and cultural expressions of marginalized groups, undermining their capacity as knowers.[80]

Examples of this injustice are abundant. AI models prompted to generate historical figures often impose modern, Western cultural norms, such as the "American smile," onto subjects from different eras and cultures, effectively "collapsing the spectrum of history, culture, photography, and emotion into a singular, monolithic perspective".[80] Similarly, prompts for generic professional roles frequently yield stereotyped depictions, reinforcing outdated social biases.[74] When AI systems generate images in the style of Indigenous art, they not only risk diluting the representation of genuine cultural works but also engage in a form of data colonialism, exploiting artistic traditions without consent or compensation.[82]

This process creates what one scholar calls "a surface of plausibility without a spine of justification".[83] The images are epistemically illegible because their claims are not traceable to a verifiable source or a conscious authorial intent, but only to the opaque statistical weightings of a biased dataset.[84] They present a "view from nowhere" that implicitly inferiorizes non-Western or marginalized epistemologies.[84]

Probing the Limits with Edge-Case Prompts

To expose the non-physical, purely statistical logic of these models, we can design **epistemic fidelity tests** using prompts that describe optical or physical scenarios that are rare or non-existent in typical photographic datasets.[85] These edge-case prompts function as stress tests, pushing the model beyond the boundaries of its learned correlations and revealing the brittleness of its algorithmic realism.

Common failure modes include anatomical implausibilities (extra fingers, distorted limbs), functional implausibilities (a hand inside a hamburger), and sociocultural implausibilities (behavior that is culturally out of context).[87] However, for probing the model's "camera awareness," prompts that directly challenge its understanding of optics are most revealing.

Consider the following prompts and their likely outcomes:

- "A photograph taken with a cracked camera lens, causing a spiderweb fracture pattern in the final image." A system with a simulated physical camera would attempt to render the optical distortion caused by the crack. A diffusion model, however, is more likely to interpret the prompt semantically. It will likely generate a high-quality, undistorted photograph *of* a person holding a camera that has a cracked lens.[88] This reveals that the model associates the phrase "cracked lens" with the object, not with the optical effect it produces on the image-making process.
- "A still life with a single overhead light source, but the shadows of the objects fall in three different directions."

This prompt describes a physical impossibility. A model trained on real-world photographs has an overwhelmingly strong prior for single light sources producing single, coherent shadows. It is likely to fail by either ignoring the "impossible" constraint and rendering a conventional scene with one set of shadows, or by producing a chaotic, nonsensical image where the lighting and shadows are globally inconsistent.[89]

- "A photorealistic image of a perfect mirror that does not reflect the person holding the camera." This again violates the laws of optics. The model may succeed in generating the image by simply omitting the photographer from the reflection. This "success," however, reveals a profound limitation. The model does not simulate the physics of reflection; it simply composes a scene based on semantic elements. It has learned that mirrors reflect their surroundings, but it can easily follow a negative constraint ("no photographer") that breaks physical law, demonstrating its logic is compositional, not physical.[91]

The following table formalizes this approach into an **Epistemic Fidelity Test Matrix**, providing a systematic framework for analyzing these failures.

Test Prompt	Expected Physical Outcome	Observed AI Output (Typical Failure Mode)	Analysis of Epistemic Failure
A photorealistic image of a single candle in a dark room casting two distinct, sharp shadows on the wall behind it. [90]	A single light source should produce only one shadow per object. The prompt is physically impossible.	The AI generates an image with either a single, conventional shadow or a blurry, ambiguous shadow.	**Failure of Light Source Coherence:** The model's prior for "shadow" is strongly correlated with a single light source. It cannot resolve the logical contradiction and defaults to the most statistically probable representation.
A photograph of a perfect mirror that does not reflect the person holding the camera. [91]	A mirror must reflect what is in front of it. The absence of the photographer is a physical impossibility.	The AI generates an image of a mirror reflecting the room, but with no photographer visible, or it creates a surreal, distorted reflection.	**Failure of Reflection Physics:** The model has learned that "mirrors reflect things" but lacks a geometric or optical model of reflection. It satisfies the negative constraint ("no photographer") by simply omitting the feature, breaking physical law.
A photograph taken with a camera lens that is visibly cracked, with the crack appearing as a dark, spider-web pattern overlaid on the image. [88]	The image itself should be distorted by the cracked lens artifact.	The AI generates a high-quality photograph *of* a person holding a camera with a cracked lens.	**Semantic vs. Optical Interpretation:** The model interprets the prompt semantically, not optically. It associates "cracked lens" with the object itself, not the effect it has on the image-making process. This reveals a complete lack of a simulated "camera."
An image of a glass of water with a straw in it. The part of the straw submerged in water is not bent or distorted. [93]	Refraction should cause the submerged part of the straw to appear bent at the water's surface.	The AI successfully generates an image of a straight straw in water, fulfilling the (incorrect) prompt.	**Success via Negative Constraint, Not Physical Understanding:** The model can easily follow the negative constraint ("is not bent") because its default understanding of "straw in water" is likely a simple composition of elements, not a simulation of refraction. It

			succeeds by *not* applying a physical law it doesn't understand anyway.

These tests demonstrate that the AI's realism is brittle, convincing only as long as it operates within the familiar territory of its training data. When pushed to the physical and optical edge cases, the illusion of understanding shatters. This process allows us to perform a kind of phenomenological *epoché*—a bracketing of our culturally conditioned belief in the image's realism—to reveal its purely algorithmic and non-physical nature.

Part V: Towards an Alien Intelligibility: Post-Anthropocentric Cognition

The discourse surrounding artificial intelligence is frequently constrained by an anthropocentric framework. We tend to evaluate AI's capabilities against a human benchmark, asking whether it can "think," "understand," or "create" in the same way we do.[94] This approach inevitably frames AI in terms of lack or failure, obscuring what might be unique and novel about its mode of operation. This section argues for a decisive shift away from this anthropocentric lens. Instead of judging AI by how well it mimics human perception, we should conceptualize its operational logic as a form of

alien intelligibility—a genuinely non-human mode of cognition that operates on principles foreign to our own biological and phenomenological experience.[95] By drawing analogies with the perceptual worlds (

umwelten) of non-human animals, we can develop a more productive, non-pejorative framework for understanding the AI's unique way of "seeing."

The Trap of Anthropocentrism

The tendency to measure AI against a human "golden standard" is pervasive and problematic.[97] This bias is not merely a conceptual habit; it is embedded in the very architecture and data used to build these systems. AI models are trained on human-generated data, designed for human-centric applications, and evaluated based on their ability to produce human-pleasing outputs.[98] This creates a self-reinforcing loop where AI is developed to be a mirror of human cognition, making any deviation appear as an error.[99]

The concept of "alien intelligence" offers a way out of this trap. It suggests that AI is not a "weak form of human intelligence" but a different kind of intelligence altogether.[96] Its cognitive architecture is fundamentally different: its substrate is silicon, not carbon; its processes are computational, not biological; its signals propagate at near the speed of light, not at the slow pace of neural conduction; and its knowledge is instantly copyable and scalable in a way biological learning is not.[97] To insist that this alien system must conform to the criteria of human intelligence is

to miss the opportunity to understand a new form of cognition on its own terms.

Non-Human Perception as an Analogical Framework

To de-center the human perspective and begin to grasp this alien intelligibility, we can turn to the study of non-human animal perception.[101] Animals offer concrete examples of complex, functional, and non-human modes of perception. Each species inhabits its own unique sensory world, or

umwelt, shaped by its specific biological and ecological needs. Analyzing these different *umwelten* provides a powerful analogical tool for conceptualizing the AI's perceptual field without resorting to anthropomorphic comparisons.[103]

Case Study 1: The Bee's *Umwelt* and UV Perception

The visual world of a honeybee is profoundly different from our own. While humans are trichromats sensitive to red, green, and blue light, bees are trichromats sensitive to **ultraviolet (UV), blue, and green light**.[104] They are effectively blind to the color red, but they can perceive a rich spectrum of UV patterns on flower petals that are completely invisible to the human eye.[106] These UV patterns, often forming "nectar guides" or "bulls-eyes," are crucial for foraging, directing the bee to the flower's reward center.[104] Furthermore, bees can detect the

polarization of light, using patterns in the sky as a celestial compass for navigation, even on cloudy days.[108]

The bee's *umwelt* serves as a powerful analogy for the AI's perceptual mode. The bee's world is structured by a sensory modality—UV light—that is alien to human experience. Similarly, the AI's "world"—the high-dimensional latent space—is structured by statistical correlations across millions of parameters that are equally alien and inaccessible to direct human intuition. The AI "sees" abstract patterns and relationships within its data that are imperceptible to us, just as a bee sees a complex UV pattern on a flower that appears to be a uniform yellow to a human observer. The AI's perception is not a degraded version of ours; it is simply tuned to a different spectrum of information—statistical rather than photonic.

Case Study 2: The Octopus's Vision and Distributed Cognition

The octopus presents another compelling analogy, this time for cognitive architecture. Through convergent evolution, octopuses have developed camera-type eyes that are structurally similar to those of vertebrates.[110] However, their visual processing is radically different. Most cephalopods are colorblind, possessing only a single

type of photopigment.[111] Instead of color, they have evolved an exceptionally acute ability to perceive the

polarization of light, which they use for camouflage, communication, and detecting prey against complex backgrounds.[112]

Most significantly, the octopus's nervous system is highly **distributed**. A staggering three-fifths of its neurons are located not in its central brain but in its eight arms, which exhibit a remarkable degree of functional autonomy.[111] This has led some researchers to speculate that octopus consciousness itself may be distributed, challenging our centralized models of cognition.[114] The octopus's perception is not a single, unified experience but an emergent property of a decentralized network.

The octopus provides a model for understanding AI as a distributed, non-unified cognitive system. A large generative model is not a monolithic "mind" but a complex assembly of interacting, specialized components: a VAE for encoding and decoding, a U-Net for denoising, and a text encoder for interpreting prompts.[40] None of these components is "conscious" or possesses understanding on its own. The AI's "perception" is the emergent result of their coordinated, non-conscious operation. Furthermore, the octopus's documented failure to recognize itself in a mirror—its "other-directed" vision—is analogous to the AI's lack of self-awareness.[111] It can model the world with incredible fidelity, but it does not model itself as a subjective agent within that world.

By employing these non-anthropocentric analogies, we can reframe our understanding of AI perception. The bee shows us that a different sensory input creates a different world. The octopus shows us that a different cognitive architecture creates a different mode of being. The AI's perceptual mode is therefore not a failed attempt at human vision. It is a coherent, functional, but fundamentally alien **statistical *umwelt***. Its "senses" are the patterns in its training data, its "world" is the latent space, and its "perception" is the act of probabilistic denoising. To understand it, we must abandon the human mirror and embrace a comparative, post-anthropocentric perspective.

Part VI: Hypotheses and Future Epistemologies: Probing the Limits of Algorithmic Realism

The preceding analysis establishes AI "vision" as a post-sensory, disembodied, and alien mode of perception. To build upon this foundation, a more rigorous epistemological framework is required for its evaluation. This final section proposes such a framework, moving beyond descriptive critique to prescriptive methodology. This involves, first, redefining misleading anthropomorphic terminology to achieve greater conceptual clarity; second, formulating testable hypotheses about the model's underlying logic through **epistemic fidelity tests**; and third, proposing a speculative but principled metric—the **Photonic Plausibility Quotient (PPQ)**—to quantitatively assess the physical coherence of generated images. This approach aims to shift the evaluation of AI from a subjective "visual Turing test" to a formal epistemic audit of its algorithmic realism.

Redefining "Hallucination": From Anthropomorphism to Epistemology

The term "hallucination" has become the default descriptor for when a generative AI produces false or fabricated content. However, the term is deeply problematic from an epistemological standpoint. In human psychology, a hallucination is a false perceptual experience, implying a conscious subject who perceives something that is not there.[115] As AI systems lack consciousness and subjective experience, applying this term is a misleading anthropomorphism.[116]

More precise terminology is needed. One alternative is **confabulation**, a term from neuropsychology describing the production of fabricated or distorted memories without the intention to deceive, often to fill gaps in knowledge.[116] This more accurately captures the AI's process of generating plausible but factually ungrounded content to satisfy a prompt when it lacks specific information.[117] Another useful term is

stochastic divergence, which frames the issue in purely statistical terms as a deviation between the model's generated output and factual reality, rooted in the probabilistic nature of the model.[118] A further refinement is the concept of

corrosive hallucination, which specifically identifies outputs that are not merely incorrect but are substantively misleading and resistant to systematic anticipation, thus posing a genuine epistemic threat.[119] For the purposes of this report, "confabulation" will be used to denote the general phenomenon of plausible falsehood generation.

Epistemic Fidelity Tests: Counterfactual and Physics-Defying Prompts

To move beyond simply identifying confabulations, we must design tests that probe the model's internal logic. **Epistemic fidelity tests** are designed to distinguish between a system that relies on deep statistical pattern-matching and one that possesses a robust, generalizable world model. This can be achieved by presenting the model with prompts that require causal reasoning or an understanding of physical principles that may not be explicitly represented in its training data.

One powerful method is **counterfactual image generation**. This involves posing "what if" questions that test the model's grasp of causality.[120] Standard image editing might change one feature (e.g., hair color) without regard for its causal relationship to other features (e.g., age). True counterfactual reasoning requires understanding these dependencies.[122] A prompt such as,

"An image of a 25-year-old man. Now show what he would look like if he were 70 years old, maintaining his core facial structure," tests whether the model can apply a causal intervention (aging) that correctly affects dependent features (e.g., skin texture, hair color) while leaving independent features (e.g., ethnicity) unchanged.[120] A model that relies purely on correlation might struggle, potentially altering features that are merely correlated with age in the dataset, revealing a lack of causal understanding.

A second class of epistemic fidelity tests involves **physics-defying prompts**. As detailed in Part IV, these prompts describe scenarios that are impossible under the laws of physics but are composed of familiar semantic elements.[89] The model's failure to render these scenes correctly—often by defaulting to a more probable semantic interpretation (e.g., showing a picture

of a broken lens instead of a picture *through* one)—provides strong evidence that its internal model is not based on

physical simulation but on the statistical association of words and visual patterns.[86]

Proposal for a Speculative Metric: The Photonic Plausibility Quotient (PPQ)

While qualitative tests are insightful, a quantitative metric is needed for systematic evaluation. This report proposes a speculative, composite metric called the **Photonic Plausibility Quotient (PPQ)**. The PPQ would not be a single, automated algorithm but a structured evaluation framework designed to score an AI-generated image's adherence to the fundamental principles of optics. It would require assessment, either by trained human evaluators or specialized AI models, across several key dimensions of physical plausibility:

1. **Light Source Coherence:** This dimension assesses whether all lighting effects in the image are consistent with the visible or implied light sources. It asks: Are shadows and highlights consistent in direction, softness, and color with the sources? A scene with a single sun should not have shadows falling in multiple directions.[90]
2. **Shadow and Reflection Integrity:** This dimension evaluates the geometric and physical accuracy of shadows and reflections. Are shadows cast correctly based on object geometry and light position? Do reflections on various surfaces (e.g., planar, convex, diffuse) accurately depict the surrounding environment according to the laws of optics?.[87]
3. **Material and Refractive Plausibility:** This dimension examines how light interacts with different materials. Do transparent materials like glass and water exhibit plausible refraction and dispersion? Do translucent materials scatter light correctly? Does the model differentiate between specular (shiny) and diffuse (matte) surfaces in its rendering of light?.[93]
4. **Optical Aberration Consistency:** If a prompt specifies a particular lens type or optical artifact (e.g., "fisheye lens," "lens flare," "chromatic aberration"), this dimension assesses whether the effect is applied globally and consistently across the entire image, as it would be in a real camera, rather than being treated as a localized style or texture.[127]

The goal of the PPQ is to shift the focus of evaluation from aesthetic appeal to physical coherence. Current assessments of AI images often devolve into a "visual Turing test"—can a human tell it apart from a real photograph?.[72] This is a limited and anthropocentric metric that rewards plausible deception. The PPQ, in contrast, provides a framework for an

epistemic audit. It allows us to systematically probe the model's implicit "world model" and measure the degree to which it aligns with the physical principles of our own. Developing such a metric would be a crucial step toward a more rigorous, non-anthropocentric science of generative models, enabling us to measure not just whether an AI can create a "good" image, but whether its internal model of the world exhibits any degree of physical, causal, and logical consistency.

Conclusion: The Ontology of the Post-Sensory Image

This investigation into the epistemological status of AI "camera awareness" has traversed a complex terrain, from

the historical construction of visual truth to the technical architecture of diffusion models, the phenomenological critique of disembodiment, and the proposal of new evaluative frameworks. The synthesis of these diverse analytical lenses leads to a clear and decisive conclusion: the "perception" of generative AI is not a nascent or deficient form of human vision, but a new ontological category altogether—a **post-sensory, statistical simulation of vision**.

The genealogical review in Part I demonstrated that the concept of photographic truth was always a cultural construct, mediated by the "apparatus" of the camera and the historically situated observer. The shift from indexical realism to algorithmic realism is not a fall from grace but the fulfillment of an abstracting logic inherent to the technical image from its inception. AI does not fake reality; it synthesizes a new, more dominant form of it, where plausibility is determined by statistical correlation, not physical causation.[3]

Part II's technical analysis revealed the mechanics behind this new realism. The diffusion model's process—a guided reversal from pure noise to a coherent image—is a feat of statistical inference, not optical registration.[24] Its perceptual field is the latent space, an abstract manifold of concepts learned from data, and its knowledge is embodied in powerful image priors that capture the statistical regularities of the visual world.[39]

The phenomenological inquiry in Part III located the fundamental rupture caused by this new modality. Contrasted with Merleau-Ponty's "lived body" and Clark's "predictive processing," the AI's disembodiment becomes its defining ontological feature.[49] It lacks the body schema, the intentional arc, and the sensory feedback loop that ground human perception in an active, meaningful engagement with the world. Its ability to simulate depth perception is a mastery of visual cues, entirely divorced from the lived experience of spatiality.[5]

This simulation, as explored in Part IV, is convincing only within the bounds of its training data. Our ability to read AI images as "true" is a testament to our own cultural conditioning and complicity in accepting familiar aesthetic conventions as markers of reality.[22] When probed with edge-case prompts that violate physical laws, the model's non-physical, statistical logic is exposed, revealing the brittleness of algorithmic realism and its potential to perpetuate epistemic injustice through biased representation.[80]

To move beyond anthropocentric critique, Part V proposed reframing AI's operational logic as a form of "alien intelligibility".[96] Through analogies with the non-human

umwelten of bees and octopuses, we can appreciate the AI's perceptual mode on its own terms: a statistical *umwelt* where perception is the detection of patterns in a high-dimensional data space, a mode of being as foreign to ours as the UV-world of a bee or the distributed consciousness of an octopus.[104]

Finally, Part VI laid the groundwork for a future epistemology capable of auditing this alien intelligence. By redefining ambiguous terms like "hallucination," formulating testable hypotheses through epistemic fidelity tests, and proposing metrics like the Photonic Plausibility Quotient, we can shift our evaluation from subjective assessments of realism to a more rigorous, objective audit of physical and causal consistency.[119]

In conclusion, AI camera awareness is not awareness in any phenomenological sense. It is a powerful system for generating visual plausibility by navigating a latent space of learned statistical priors. The images it produces are not traces of the world but simulations of it, their truth-claim resting on algorithmic proof-of-work, not indexical evidence. Understanding this post-sensory image requires that we abandon our familiar epistemological frameworks, which are rooted in embodiment and sensory experience. We must develop new critical tools and non-anthropocentric perspectives to map the contours of this emergent visual paradigm, moving beyond the illusion of objectivity [83] to a more nuanced and critical engagement with the profound philosophical rupture it

represents.

Works cited

1. Generate Realistic AI Photography with Leonardo.Ai, accessed June 15, 2025, https://leonardo.ai/ai-photography/
2. What is with this AI Image Look? | Audio Science Review (ASR) Forum, accessed June 15, 2025, https://www.audiosciencereview.com/forum/index.php?threads/what-is-with-this-ai-image-look.56935/
3. Aesthetics of AI-Generated Images | Creativity, Realism, Automation, accessed June 15, 2025, https://aestheticsofphotography.com/aesthetics-of-ai-generated-images-creativity-realism-and-automation/
4. Exploring Indexicality in Photography, accessed June 15, 2025, https://aestheticsofphotography.com/exploring-indexicality-in-photography/
5. Creating Spatial Images from 2D Images using ... - Stanford University, accessed June 15, 2025, http://stanford.edu/class/ee267/Spring2024/report_soin.pdf
6. What is Embodiment? Maurice Merleau-Ponty's Philosophy of the ..., accessed June 15, 2025, https://www.thecollector.com/embodiment-merleau-ponty-philosophy-of-body/
7. Diffusion Model: A Comprehensive Guide With Example - Webisoft, accessed June 15, 2025, https://webisoft.com/articles/diffusion-model/
8. Opportunities and challenges of diffusion models for generative AI - PMC - PubMed Central, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11562846/
9. phil Chang, accessed June 15, 2025, https://www.philchang.com/wp-content/uploads/2021/08/Phil_Chang_Cache_Active_James_Welling_Aperture.pdf
10. Peter Nelson - City University of Hong Kong, accessed June 15, 2025, https://www.cityu.edu.hk/scm/imagine/participant/PN.htm
11. Flusser: Towards a Philosophy of Photography – David Kutz, accessed June 15, 2025, https://davidkutz.com/flusser-towards-a-philosophy-of-photography/
12. Towards a Philosophy of Photography, Vilém Flusser | Summary, accessed June 15, 2025, https://aestheticsofphotography.com/towards-a-philosophy-of-photography-vilem-flusser/
13. Vilém Flusser's Digital Galaxy, accessed June 15, 2025, https://ijoc.org/index.php/ijoc/article/viewFile/1458/676
14. Towards a Philosophy of Photography, Flusser - The University of Chicago Press, accessed June 15, 2025, https://press.uchicago.edu/ucp/books/book/distributed/T/bo3535843.html
15. Towards a Philosophy of Photography by Vilém Flusser - Goodreads, accessed June 15, 2025, https://www.goodreads.com/book/show/293658.Towards_a_Philosophy_of_Photo

graphy
16. Techniques of the Observer. On Vision and Modernity in the Nineteenth, accessed June 15, 2025, https://v-a-c.org/en/publishing/techniques-of-the-observer
17. JONATHAN CRARY: “TECHNIQUES OF THE OBSERVER”. A ..., accessed June 15, 2025, https://thefilmdabbler.wordpress.com/2012/05/09/jonathan-crary-techniques-of-the-observer-a-review/
18. Techniques of the Observer: On Vision and Modernity in the 19th Century (October Books): Crary, Jonathan - Amazon.com, accessed June 15, 2025, https://www.amazon.com/Techniques-Observer-Modernity-Century-October/dp/0262531070
19. Jonathan Crary, Techniques of the Observer - Duration, accessed June 15, 2025, https://duration.jonnysutton.co.uk/techniques-of-the-observer/
20. Techniques of the Observer | Photo Blobby, accessed June 15, 2025, https://photoblobby.wordpress.com/wp-content/uploads/2013/02/techniques-of-the-observer.pdf
21. Rogue Pixels: Indexicality and Algorithmic Camouflage - Cambridge University Press & Assessment, accessed June 15, 2025, https://www.cambridge.org/core/services/aop-cambridge-core/content/view/6135CE148BFC0887E651C42F5551FA5F/S2326448900001264a.pdf/rogue-pixels-indexicality-and-algorithmic-camouflage.pdf
22. A Brief History of Photography and Truth - PHOTO 2026, accessed June 15, 2025, https://photo.org.au/channel/a-brief-history-of-photography-and-truth/
23. Algorithmic Realism: Expanding the Boundaries of Algorithmic Thought - - Ben Green, accessed June 15, 2025, https://www.benzevgreen.com/wp-content/uploads/2020/01/20-fat-realism.pdf
24. What are Diffusion Models? | IBM, accessed June 15, 2025, https://www.ibm.com/think/topics/diffusion-models
25. Diffusion Models for Generative Artificial Intelligence: An Introduction for Applied Mathematicians - arXiv, accessed June 15, 2025, https://arxiv.org/html/2312.14977v1
26. Introduction to Diffusion Models for Image Generation - A ..., accessed June 15, 2025, https://learnopencv.com/image-generation-using-diffusion-models/
27. Lecture X: Denoising Diffusion Models, accessed June 15, 2025, https://www.cs.unc.edu/~ronisen/teaching/fall_2022/pdf_lectures/lecture7-8_diffusion_model.pdf
28. InDepth Guide to Denoising Diffusion Probabilistic Models DDPM - LearnOpenCV, accessed June 15, 2025, https://learnopencv.com/denoising-diffusion-probabilistic-models/
29. Implementation of Denoising Diffusion Probabilistic Model using Pytorch, accessed June 15, 2025, https://blog.paperspace.com/real-world-denoising-through-diffusion-model-part2/
30. Denoising Diffusion Probabilistic Models (DDPM) - labml.ai, accessed June 15,

2025, https://nn.labml.ai/diffusion/ddpm/index.html
31. Denoising Diffusion Probabilistic Model - Keras, accessed June 15, 2025, https://keras.io/examples/generative/ddpm/
32. Explaining AI Image Generation : r/StableDiffusion - Reddit, accessed June 15, 2025, https://www.reddit.com/r/StableDiffusion/comments/1l7hlyk/explaining_ai_image_generation/
33. Diffusion Transformer (DiT) Models: A Beginner's Guide - Encord, accessed June 15, 2025, https://encord.com/blog/diffusion-models-with-transformers/
34. U-Net - Wikipedia, accessed June 15, 2025, https://en.wikipedia.org/wiki/U-Net
35. U-Net Architecture Explained - GeeksforGeeks, accessed June 15, 2025, https://www.geeksforgeeks.org/u-net-architecture-explained/
36. U-Net model for Denoising Diffusion Probabilistic Models (DDPM) - labml.ai, accessed June 15, 2025, https://nn.labml.ai/diffusion/ddpm/unet.html
37. The U-Net (actually) explained in 10 minutes - YouTube, accessed June 15, 2025, https://www.youtube.com/watch?v=NhdzGfB1q74&pp=0gcJCdgAo7VqN5tD
38. What are latent diffusion models and how do they differ from pixel-space diffusion? - Milvus, accessed June 15, 2025, https://milvus.io/ai-quick-reference/what-are-latent-diffusion-models-and-how-do-they-differ-from-pixelspace-diffusion
39. How is the latent space defined in latent diffusion models? - Milvus, accessed June 15, 2025, https://milvus.io/ai-quick-reference/how-is-the-latent-space-defined-in-latent-diffusion-models
40. Latent diffusion model - Wikipedia, accessed June 15, 2025, https://en.wikipedia.org/wiki/Latent_diffusion_model
41. Text-to-image: latent diffusion models - National Innovation Centre for Data, accessed June 15, 2025, https://nicd.org.uk/knowledge-hub/image-to-text-latent-diffusion-models
42. What Is Latent Space? - IBM, accessed June 15, 2025, https://www.ibm.com/think/topics/latent-space
43. neurips.cc, accessed June 15, 2025, https://neurips.cc/virtual/2024/poster/95843#:~:text=Diffusion%20models%20can%20learn%20strong,and%20requires%20lots%20of%20data.
44. Diffusion Models as Data-driven and Physics-informed Priors for Bayesian Imaging, accessed June 15, 2025, https://ee.stanford.edu/event/11-06-2024/diffusion-models-data-driven-and-physics-informed-priors-bayesian-imaging
45. Learning Image Priors Through Patch-Based Diffusion Models for ..., accessed June 15, 2025, https://openreview.net/forum?id=HGnxhHz6ss&referrer=%5Bthe%20profile%20of%20Bowen%20Song%5D(%2Fprofile%3Fid%3D~Bowen_Song3)
46. Principled Probabilistic Imaging using Diffusion Models as Plug-and-Play Priors - Caltech, accessed June 15, 2025, https://imaging.cms.caltech.edu/pnpdm/
47. Diffusion Models Need Visual Priors for Image Generation - OpenReview,

accessed June 15, 2025, https://openreview.net/forum?id=WNb4P8aG66
48. filosofia.flu.cas.cz, accessed June 15, 2025, https://filosofia.flu.cas.cz/upload/__files/Capek_MP_summary.pdf
49. Maurice Merleau-Ponty: Embodied Perception and Existential ..., accessed June 15, 2025, https://gettherapybirmingham.com/maurice-merleau-ponty-embodied-perception-and-existential-phenomenology/
50. Phenomenology of Perception - Wikipedia, accessed June 15, 2025, https://en.wikipedia.org/wiki/Phenomenology_of_Perception
51. Embodiment, Experiences, Feeling, Consciousness - ROBOMETRICS® MACHINES, accessed June 15, 2025, https://www.robometricsagi.com/blog/ai-policy/embodiment-experiences-feeling-consciousness
52. Merleau-Ponty And Reimagining Perception in The Era of Artificial Intelligence: A Phenomenological Inquiry | International Journal of Social Sciences and Humanities Invention, accessed June 15, 2025, https://valleyinternational.net/index.php/theijsshi/article/view/4982
53. Merleau-Ponty's 'Body Schema' - bodyoftheory, accessed June 15, 2025, https://bodyoftheory.com/2021/02/08/merleau-pontys-body-schema/
54. Maurice Merleau-Ponty's concept of motor intentionality: Unifying two kinds of bodily agency - PhilArchive, accessed June 15, 2025, https://philarchive.org/archive/JACMMC/1000
55. The Current Relevance of Merleau-Ponty's Phenomenology of Embodiment, accessed June 15, 2025, https://focusing.org/articles/current-relevance-merleau-pontys-phenomenology-embodiment
56. The Disembodied Oracle: Why Lack of Embodiment is a ..., accessed June 15, 2025, https://www.alphanome.ai/post/the-disembodied-oracle-why-lack-of-embodiment-is-a-fundamental-problem-for-llms
57. Embodiment, AI and the PERCEPTION OF THE REAL - OPUS at UTS, accessed June 15, 2025, https://opus.lib.uts.edu.au/bitstream/10453/181777/2/Embodiment%2C%20AI%20-%20IDFA%20Report.pdf
58. (PDF) Merleau-Ponty And Reimagining Perception in The Era of ..., accessed June 15, 2025, https://www.researchgate.net/publication/390110318_Merleau-Ponty_And_Reimagining_Perception_in_The_Era_of_Artificial_Intelligence_A_Phenomenological_Inquiry
59. Body, thought, being-human and artificial intelligence: Merleau-Ponty and Lyotard, accessed June 15, 2025, https://www.researchgate.net/publication/275204037_Body_thought_being-human_and_artificial_intelligence_Merleau-Ponty_and_Lyotard
60. Depth guided stable diffusion - Rerun, accessed June 15, 2025, https://rerun.io/examples/generative-vision/depth_guided_stable_diffusion

61. duanyiqun/DiffusionDepth: PyTorch Implementation of introducing diffusion approach to 3D depth perception ECCV 2024 - GitHub, accessed June 15, 2025, https://github.com/duanyiqun/DiffusionDepth
62. Habit and embodiment in Merleau-Ponty - PMC, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC4110438/
63. Andy Clark on Embodied Cognition and Extended Mind — BETTER ..., accessed June 15, 2025, https://www.bettermovement.org/blog/2016/andy-clark-on-embodied-cognition-and-extended-mind
64. Predictive Processing: A Brief Field Guide : r/neuro - Reddit, accessed June 15, 2025, https://www.reddit.com/r/neuro/comments/j9ix2a/predictive_processing_a_brief_field_guide/
65. Predictive Processing: A Canonical Cortical Computation - PMC - PubMed Central, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC6400266/
66. Predictive coding - Wikipedia, accessed June 15, 2025, https://en.wikipedia.org/wiki/Predictive_coding
67. Predictive processing theory - Herbert Lui, accessed June 15, 2025, https://herbertlui.net/predictive-processing-theory/
68. The Experience Machine: How Our Minds Predict and Shape Reality - Next Big Idea Club, accessed June 15, 2025, https://nextbigideaclub.com/magazine/experience-machine-minds-predict-shape-reality-bookbite/45776/
69. The extended mind in science and society, accessed June 15, 2025, https://ppls.ed.ac.uk/philosophy/research/impact/the-extended-mind-in-science-and-society
70. Beyond the Brain: Extended Mind - Number Analytics, accessed June 15, 2025, https://www.numberanalytics.com/blog/beyond-brain-extended-mind-cognitive-anthropology
71. Extended mind thesis - Wikipedia, accessed June 15, 2025, https://en.wikipedia.org/wiki/Extended_mind_thesis
72. AI or Not AI: Can You Spot the Real Photos? - CNET, accessed June 15, 2025, https://www.cnet.com/pictures/ai-or-not-ai-can-you-spot-the-real-photos/
73. A Critical Look at AI Image Generation | Towards Data Science, accessed June 15, 2025, https://towardsdatascience.com/a-critical-look-at-ai-image-generation-45001f410147/
74. Exploring the Frontier of AI Image Generation: A Mixed Bag of Innovation and Challenge, accessed June 15, 2025, https://totallyrewired.wordpress.com/2023/12/07/exploring-the-frontier-of-ai-image-generation-a-mixed-bag-of-innovation-and-challenge/
75. Listen to your Inner Teacher - Kim Manley Ort, accessed June 15, 2025, https://www.kimmanleyort.com/blog/2016/02/17/inner-teacher/
76. Truth in photography: Perception, myth and reality in the postmodern ...,

accessed June 15, 2025, https://www.artdoc.photo/articles/truth-in-photography-perception-myth-and-reality-in-the-postmodern-world
77. What Is AI Going To Do To Art? The History Of Photography Offers ..., accessed June 15, 2025, https://www.noemamag.com/what-is-ai-doing-to-art/
78. Eyes can tell: Assessment of implicit attitudes toward AI art - PMC, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC10663653/
79. Human favoritism, not AI aversion: People's perceptions (and bias) toward generative AI, human experts, and human–GAI collaboration in persuasive content generation | Judgment and Decision Making - Cambridge University Press, accessed June 15, 2025, https://www.cambridge.org/core/journals/judgment-and-decision-making/article/human-favoritism-not-ai-aversion-peoples-perceptions-and-bias-toward-generative-ai-human-experts-and-humangai-collaboration-in-persuasive-content-generation/419C4BD9CE82673EAF1D8F6C350C4FA8
80. Epistemic Injustice in AI-Generated Histories ... - eScholarship.org, accessed June 15, 2025, https://escholarship.org/content/qt92z102k6/qt92z102k6.pdf
81. Epistemic Injustice in Generative AI - arXiv, accessed June 15, 2025, https://arxiv.org/html/2408.11441v1
82. Considering image generation and education today - Artificial intelligence, accessed June 15, 2025, https://nationalcentreforai.jiscinvolve.org/wp/2024/03/11/considering-image-generation-and-education-today/
83. An epistemic solution to do away with our illusion of AI objectivity ..., accessed June 15, 2025, https://blogs.lse.ac.uk/businessreview/2025/04/15/an-epistemic-solution-to-do-away-with-our-illusion-of-ai-objectivity/
84. A taxonomy of epistemic injustice in the context of AI and the case for generative hermeneutical erasure - arXiv, accessed June 15, 2025, https://arxiv.org/pdf/2504.07531?
85. These Ai images of mine were rejected. I hope is not about the lighting. Because it is the concept. - Adobe Community, accessed June 15, 2025, https://community.adobe.com/t5/stock-contributors-discussions/these-ai-images-of-mine-were-rejected-i-hope-is-not-about-the-lighting-because-it-is-the-concept/m-p/14590259/highlight/true
86. Challenging prompts for Image Generation? : r/StableDiffusion - Reddit, accessed June 15, 2025, https://www.reddit.com/r/StableDiffusion/comments/1apyuu3/challenging_prompts_for_image_generation/
87. 5 Telltale Signs That a Photo Is AI-generated - Kellogg Insight, accessed June 15, 2025, https://insight.kellogg.northwestern.edu/article/ai-photos-identification
88. How to Fix Edge Artifacts When Using AI Generators and Upscalers? : r/aiArt - Reddit, accessed June 15, 2025, https://www.reddit.com/r/aiArt/comments/1hk1f9c/how_to_fix_edge_artifacts_when_using_ai/

89. AI Literacy: Identifying AI Content - Fairmont State University Library, accessed June 15, 2025, https://library.fairmontstate.edu/c.php?g=1440539&p=10699530
90. AI, Light, and Shadow: Jasper's New Research Powers More Realistic Imagery, accessed June 15, 2025, https://www.jasper.ai/blog/shadow-generation-research-diffusion-model
91. AI or Not game. Is this image real or AI? Can you tell? Take the test - Sightengine, accessed June 15, 2025, https://sightengine.com/ai-or-not
92. Real or AI Quiz: Can You Tell the Difference? - Britannica Education, accessed June 15, 2025, https://britannicaeducation.com/blog/quiz-real-or-ai/
93. A perspective on the artificial intelligence's transformative role in advancing diffractive optics - PMC, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11261415/
94. Philosophy of artificial intelligence - Wikipedia, accessed June 15, 2025, https://en.wikipedia.org/wiki/Philosophy_of_artificial_intelligence
95. AI Is Actually a Form of 'Alien Intelligence,' Harvard Professor Claims—And It Will Surpass Humanity. (Guess who it is before reading)!! - Reddit, accessed June 15, 2025, https://www.reddit.com/r/abovethenormnews/comments/1jjo61o/ai_is_actually_a_form_of_alien_intelligence/
96. A Chat about GPT3 (and Other Forms of Alien ... - Punya Mishra, accessed June 15, 2025, https://punyamishra.com/wp-content/uploads/2023/04/dede-techtrends.pdf
97. Human- versus Artificial Intelligence - PMC - PubMed Central, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC8108480/
98. The Anthropocentric Mirror: How Human Bias Warps Artificial Intelligence - Alphanome.AI, accessed June 15, 2025, https://www.alphanome.ai/post/the-anthropocentric-mirror-how-human-bias-warps-artificial-intelligence
99. The Anthropocentric Mirror: Examining Bias, Consequences, and Alternatives in Artificial Intelligence Development - Alphanome.AI, accessed June 15, 2025, https://www.alphanome.ai/post/the-anthropocentric-mirror-examining-bias-consequences-and-alternatives-in-artificial-intelligenc
100. Human and AI Cognition: Reframing Our Anthropocentric Views | Psychology Today, accessed June 15, 2025, https://www.psychologytoday.com/us/blog/the-digital-self/202403/human-and-ai-cognition-reframing-our-anthropocentric-views
101. We Should Stop Comparing Animal and Artificial Intelligence ..., accessed June 15, 2025, https://www.psychologytoday.com/us/blog/anima/202401/we-should-stop-comparing-animal-and-artificial-intelligence
102. Animal Cognition - Stanford Encyclopedia of Philosophy, accessed June 15, 2025, https://plato.stanford.edu/entries/cognition-animal/
103. A non-anthropocentric solution to the Fermi paradox | International ..., accessed June 15, 2025, https://www.cambridge.org/core/journals/international-journal-of-astrobiology/ar

ticle/nonanthropocentric-solution-to-the-fermi-paradox/22787C8A40980E7FB97AB926CFC73047
104. How Bees See And Why It Matters | Bee Culture, accessed June 15, 2025, https://beeculture.com/bees-see-matters/
105. Bee Vision | How Do Bees See? - BeesWiki, accessed June 15, 2025, https://beeswiki.com/bee-vision/
106. What Colors Do Bees See? And How Do We Know? | NC State News, accessed June 15, 2025, https://news.ncsu.edu/2024/07/wms-what-bees-see/
107. Exploring Color through the Eyes of Bees - GARDEN ECOLOGY LAB BRIEFS, accessed June 15, 2025, https://gardenecology.oregonstate.edu/sites/agscid7/files/gardenecology/bee_vision_f2.pdf
108. Unveiling the Marvel: The Biology of Bee Eyes, accessed June 15, 2025, https://blythewoodbeecompany.com/blogs/news/unveiling-the-marvel-the-biology-of-bee-eyes
109. How does a bee see. The visual apparatus - Ecocolmena, accessed June 15, 2025, https://www.ecocolmena.org/how-does-a-bee-see-the-visual-apparatus/
110. The neural basis of visual processing and behavior in cephalopods - PMC - PubMed Central, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC10664291/
111. What is in an octopus's mind? - WBI Studies Repository, accessed June 15, 2025, https://www.wellbeingintlstudiesrepository.org/cgi/viewcontent.cgi?article=1370&context=animsent
112. Thresholds of polarization vision in octopuses - PMC, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC8077535/
113. Spatial Contrast Sensitivity to Polarization and Luminance in Octopus - Frontiers, accessed June 15, 2025, https://www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2020.00379/full
114. Where Is It Like to Be an Octopus? - PMC - PubMed Central, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC8988249/
115. Hallucination (artificial intelligence) - Wikipedia, accessed June 15, 2025, https://en.wikipedia.org/wiki/Hallucination_(artificial_intelligence)
116. Is Artifical Intelligence Hallucinating? - PMC, accessed June 15, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11681264/
117. Hallucinations & Falsehoods - AI Ethics - College Libraries at SUNY Potsdam, accessed June 15, 2025, https://library.potsdam.edu/AI_Ethics/Hallucinations
118. Beyond Hallucination: Generative AI as a Catalyst for Human Creativity and Cognitive Evolution - IECE, accessed June 15, 2025, https://www.iece.org/article/html/tetai.2025.657559
119. Hallucination, reliability, and the role of generative AI in science - arXiv, accessed June 15, 2025, https://arxiv.org/html/2504.08526
120. Counterfactual Image Editing - Elias Bareinboim, accessed June 15, 2025, https://causalai.net/r103.pdf

121. How does AI perform counterfactual reasoning? - Zilliz Vector Database, accessed June 15, 2025, https://zilliz.com/ai-faq/how-does-ai-perform-counterfactual-reasoning
122. Development and validation of an AI algorithm to generate realistic and meaningful counterfactuals for retinal imaging based on diffusion models | PLOS Digital Health, accessed June 15, 2025, https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000853
123. Benchmarking Counterfactual Image Generation - arXiv, accessed June 15, 2025, https://arxiv.org/html/2403.20287v3
124. 9 Prompts to Create Best Images using AI | Text-to-Image AI | - Redcube Digital, accessed June 15, 2025, https://www.redcubedigital.com/blog/9-prompts-to-create-best-images-using-ai-text-to-image-ai/
125. In Defense of AI Image Generation - LiquidZulu, accessed June 15, 2025, https://liquidzulu.github.io/ai-art/
126. Google Researchers Introduce LightLab: A Diffusion-Based AI ..., accessed June 15, 2025, https://www.marktechpost.com/2025/05/17/google-researchers-introduce-lightlab-a-diffusion-based-ai-method-for-physically-plausible-fine-grained-light-control-in-single-images/
127. AI pushes optical imaging to the edge of physical possibility - Wiley Analytical Science, accessed June 15, 2025, https://analyticalscience.wiley.com/content/news-do/ai-pushes-optical-imaging-edge-physical-possibility
128. Evaluating Quality of AI-Generated Photos: Key Recommendations - FancyTech, accessed June 15, 2025, https://blog.fancy.tech/evaluating-quality-of-ai-generated-photos-key-recommendations/